



**2020**

TRIAL – YEAR 12

HIGHER SCHOOL CERTIFICATE EXAMINATION

# Mathematics Advanced

---

## General

## Instructions

- Reading time – 10 minutes
- Working time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided at the back of this paper
- In Questions 11-16, show relevant mathematical reasoning and/or calculations

---

## Total marks:

**100**

## Section I – 10 marks (pages 2-5)

- Attempt Questions 1-10
- Allow about 15 minutes for this section

## Section II – 90 marks (pages 6-34)

- Attempt Questions 11-16
- Allow about 2 hours and 45 minutes for this section

**Section I - 10 marks****Allow 15 minutes for this section**

1. Which expression is equal to  $\int \tan^2 x \, dx$ ?

(A)  $\frac{\tan^3 x}{3} + C$

(B)  $\tan x - x + C$

(C)  $\tan x + x + C$

(D)  $\sec^2 x + C$

2.  $\frac{d}{dx} \log_e \frac{4x^2 - 9}{2x - 3}$  is equal to which of the following?

(A)  $\frac{6}{2x - 3}$

(B)  $\frac{2}{2x + 3}$

(C)  $\frac{6(2x + 3)}{(2x - 3)^2}$

(D)  $\frac{6(4x + 1)}{(2x - 3)^2}$

3. Which of the following could be a primitive for  $f'(x) = \frac{x}{e^{x^2 - 8}}$  ?

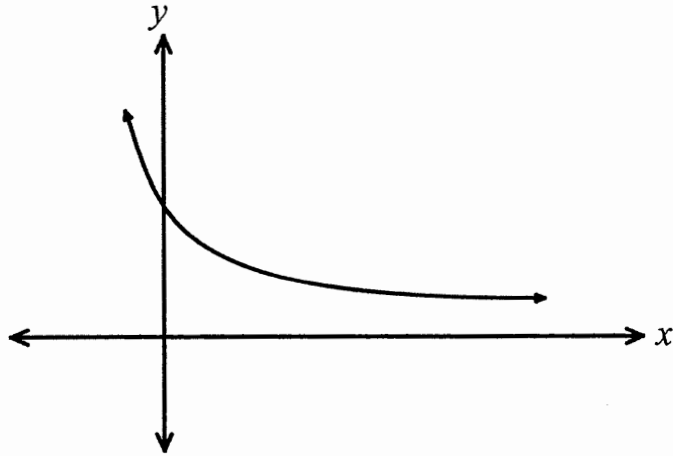
(A)  $-\frac{1}{2}(e^{x^2 - 8}) + 8$

(B)  $\frac{1}{2} \ln(e^{x^2 - 8}) + 8$

(C)  $\ln(e^{8 - x^2}) - 8$

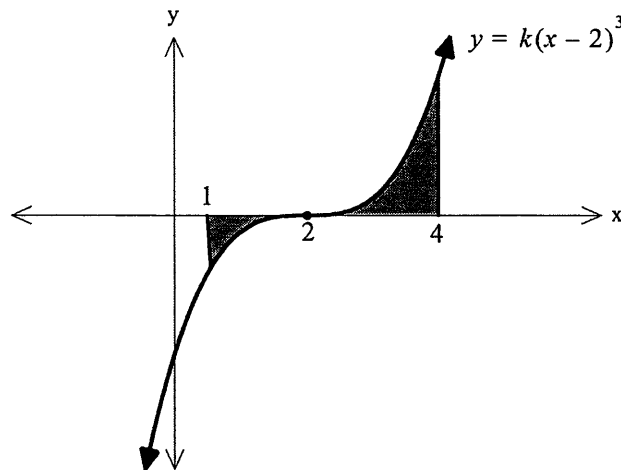
(D)  $-\frac{1}{2}(e^{8 - x^2}) - 8$

4. For the curve shown, which inequalities are correct?



- (A)  $\frac{dy}{dx} > 0$  and  $\frac{d^2y}{dx^2} > 0$
- (B)  $\frac{dy}{dx} > 0$  and  $\frac{d^2y}{dx^2} < 0$
- (C)  $\frac{dy}{dx} < 0$  and  $\frac{d^2y}{dx^2} < 0$
- (D)  $\frac{dy}{dx} < 0$  and  $\frac{d^2y}{dx^2} > 0$
5. Results for a test are given as z-scores. In this test Angela gained a z- score of 3. The test has a mean of 55 and standard deviation of 6. What was Angela's actual mark in this test?
- (A) 58
- (B) 73
- (C) 64
- (D) 67

6. The graph with the equation  $y = k(x - 2)^3$  is shown below, for some positive constant  $k$ .



If the area of the shaded region is 34, what is the value of  $k$ ?

- (A)  $\frac{136}{15}$
- (B) 8
- (C) 4
- (D)  $\frac{34}{9}$
7. The time,  $T$ , in seconds that divers can hold their breath is normally distributed with  $\mu = 120$  and  $Var(T) = 400$ . In what range of time length would you expect to find the middle 95%?
- (A)  $100 \leq x \leq 140$
- (B)  $80 \leq x \leq 160$
- (C)  $60 \leq x \leq 180$
- (D)  $40 \leq x \leq 200$

8. The exact value of  $I = \int_1^2 \frac{\ln x}{x} dx = \frac{1}{2}(\ln 2)^2$ . The approximation of  $I$  using the Trapezoidal Rule with 2 function values is
- (A) smaller by 28%
  - (B) larger by 28%
  - (C) smaller by 72%
  - (D) larger by 72%
9. Given a function  $f(x) = \frac{x}{x^2 - 5}$
- Which of the following statements is true?
- (A)  $f(x)$  is even and one-to-one.
  - (B)  $f(x)$  is even and many-to-one.
  - (C)  $f(x)$  is odd and one-to-one.
  - (D)  $f(x)$  is odd and many-to-one.
10. The amount  $M$  of certain medicine present in the blood after  $t$  hours is given by  $M = 9t^2 - t^3$  for  $0 \leq t \leq 9$ .
- When is the amount of medicine in the blood increasing most rapidly?
- (A)  $t = 0$
  - (B)  $t = 9$
  - (C)  $t = 6$
  - (D)  $t = 3$

**END OF SECTION I**

## Section II- Extended Response

Attempt Questions 11-16.

Allow about 2 hours and 45 minutes for this section.

### Question 11(15 Marks)

a) Differentiate the following

(i)  $y = (4x - 5)(4x + 5)$

1

.....

.....

.....

.....

.....

(ii)  $y = \sin^2 x$

2

.....

.....

.....

.....

.....

b) In an arithmetic series, the third term is 5 and the tenth term is 26. Find the sum of the first 14 terms. 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

Question 11 continued on the next page

c) Evaluate

2

$$\int_1^4 5(9x - 4)^4 dx$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

d) Solve the following equation for  $x$ .

2

$$e^{2x} + 3e^x - 10 = 0.$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 11 continued on the next page**

- e) (i) Show that  $\frac{d}{dx}(\sec^2 x) = 2 \tan x \sec^2 x$ . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Hence find  $\int \tan x \sec^2 x \, dx$ . 1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 11 continued on the next page**

f) Given a function  $f(x) = \begin{cases} 6x - 6x^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$

(i) Show that  $f(x)$  represents probability density function.

2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Find the mode of the probability density function  $f(x)$ .

1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**End of Question 11**

### Question 12 (13 Marks)

- a) Find the value(s) of  $b$  such that  $y = 2x + b$  is a tangent to the parabola 2

$$y = 2x^2 + 6x - 5.$$

[illegible]

**Question 12 continued on the next page**

- b) Angela guesses three questions in her multiple choice test, which has four options per question. Find the probability that Angela gets:

(i) Only one correct answer.

1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) At least one correct answer.

1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

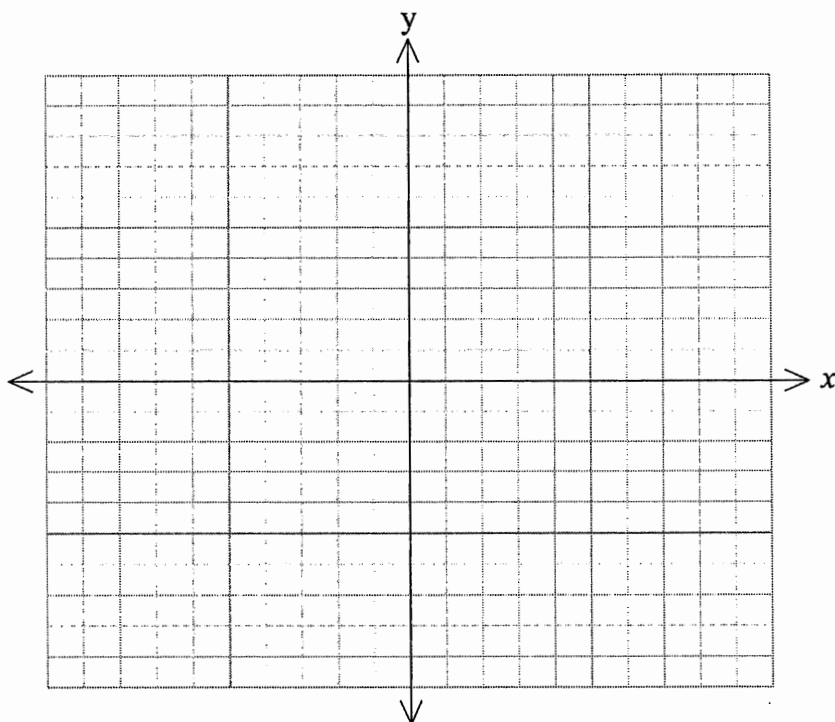
.....

.....

**Question 12 continued on the next page**

c)

- (i) Sketch the hyperbola by shifting  $y = \frac{1}{x-1}$  horizontally 3 units to the right 2 and 1 unit down.



- (ii) State the equation of the shifted hyperbola, then find all the intercepts of the shifted hyperbola with the axes and mark them on your graph in part (i). 2

.....

.....

.....

.....

.....

.....

.....

**Question 12 continued on the next page**

d) Consider the piece -wise defined function.

$$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ 4 - x^2 & x > 1 \end{cases}$$

(i) Find  $f(1)$

1

.....

.....

(ii) Find  $x$  if  $f(x) = 0$

2

.....

.....

.....

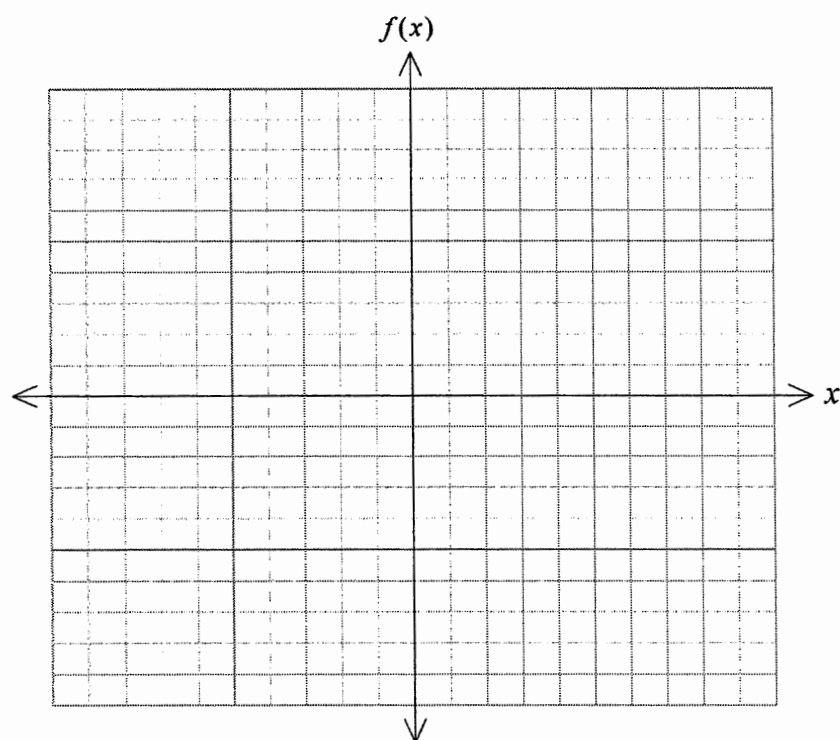
.....

.....

.....

(iii) Sketch the function showing all intercepts.

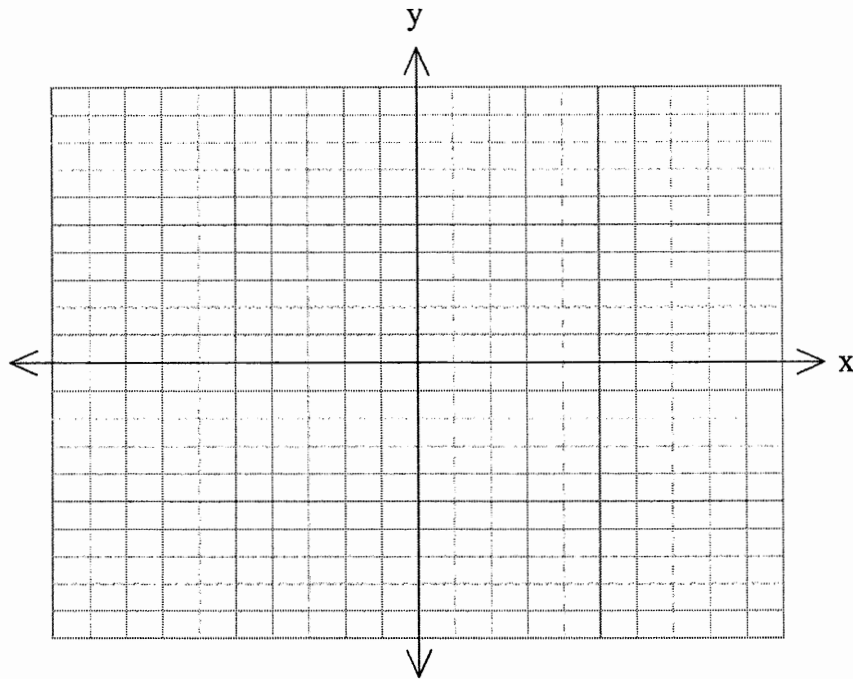
2



**End of Question 12**

**Question 13 (18 Marks)**

- a) (i) Sketch the graphs of  $f(x) = 2x - 2x^2$  and  $g(x) = x - 1$  on the same number plane. 2



- (ii) Using your graphs from part (i), or otherwise solve the inequality 2

$$x - 1 < 2x - 2x^2.$$

.....

.....

.....

.....

.....

.....

.....

.....

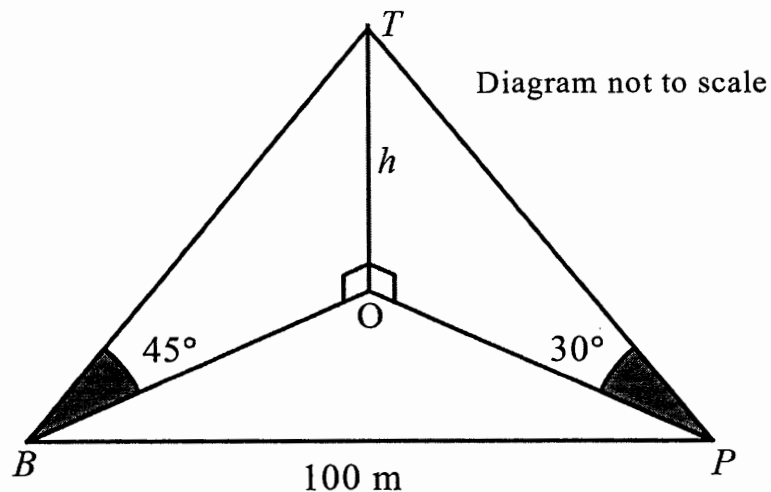
.....

.....

.....

**Question 13 continued on the next page**

- b) A surveyor stands at a point  $P$ , which is due east of the tower  $OT$ , of height  $h$  metres. The angle of elevation of the top of the tower  $T$  from  $P$  is  $30^\circ$ . The surveyor then walks 100 metres to point  $B$ , which is on a bearing of  $150^\circ$  from the foot of tower  $O$ . The angle of elevation of the top of the tower from  $B$  is now  $45^\circ$ .



- (i) Express the length of  $OP$  in terms of  $h$ .

**1**

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the entire width of the page. There are no margins, text, or other markings present.

**Question 13 continued on the next page**

(ii) Show that  $(100)^2 = h^2 + \frac{h^2}{\tan^2 30^\circ} - \frac{h^2}{\tan 30^\circ}$  . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(iii) Hence find the height of the tower. Answer correct to 1 decimal place. 1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 13 continued on the next page**

- c) The following information shows a group of people's waist measurements and weights.

|                    |    |    |    |    |    |    |    |     |
|--------------------|----|----|----|----|----|----|----|-----|
| Waist<br>(cm) $x$  | 72 | 67 | 85 | 96 | 80 | 90 | 98 | 105 |
| Weight<br>(kg) $y$ | 58 | 50 | 72 | 85 | 70 | 79 | 82 | 84  |

- (i) Calculate the correlation coefficient,  $r$ , for their waist and weight measurements 2 correct to 3 decimal places and hence describe the strength of the relationship.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Find the equation of the Least -Squares Regression Line. 1

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 13 continued on the next page**



(iii) Find any point(s) of inflection.

2

.....

.....

.....

.....

.....

.....

.....

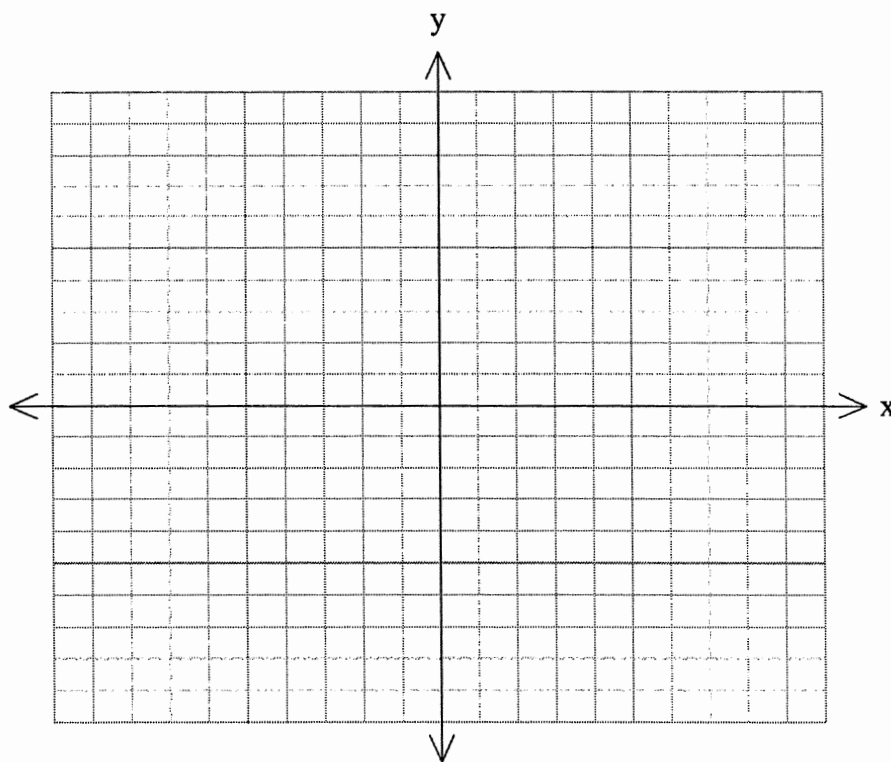
.....

.....

.....

(iv) Sketch the graph of  $f(x) = \ln(x^2 + 1)$  showing all features from part (ii) and (iii).

2



**End of Question 13**

**Question 14 (14 marks)**

- a) (i) Prove the following identity

**1**

$$(1 + \tan x)^2 = 2 \tan x + \sec^2 x$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Hence find the area bounded by  $y = (1 + \tan x)^2$  and the  $x$ -axis between **3**

$$-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}.$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 14 continued on the next page**

b) Given  $y = 2\sin\left(2x - \frac{\pi}{3}\right)$

(i) State the amplitude and period.

2

.....

.....

.....

.....

.....

.....

.....

(ii) Find the exact values of all intercepts of

2

$y = 2\sin\left(2x - \frac{\pi}{3}\right)$  with the axes for  $0 \leq x \leq \pi$ .

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

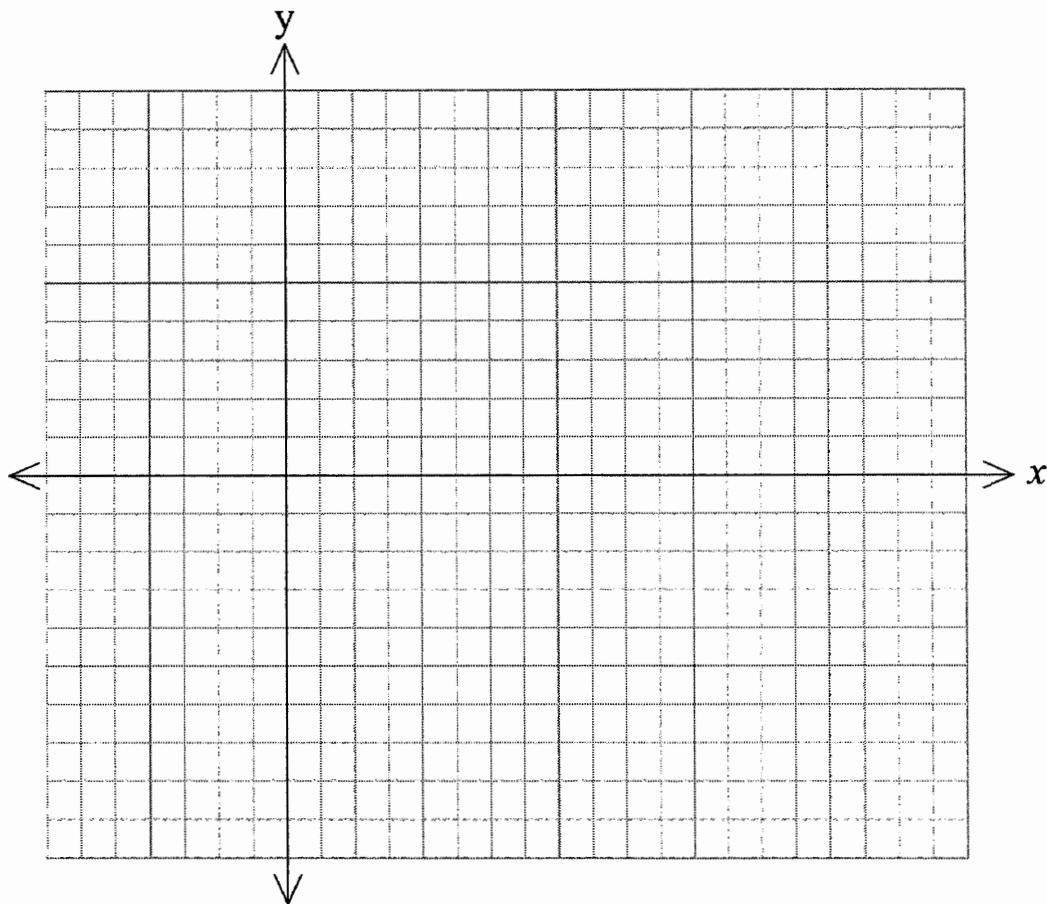
.....

.....

.....

**Question 14 continued on the next page**

- (iii) Hence sketch the graph of  $y = 2\sin\left(2x - \frac{\pi}{3}\right)$  for  $0 \leq x \leq \pi$ , 2  
showing all features from part (i) and (ii) and the global maximum and minimum.



**Question 14 continued on the next page**

c) A bag contains three red balls and four black balls. Two balls are selected at random without replacement from the bag.

Let  $X$  be the number of black balls drawn.

- (i) Fill in the following table and hence find exact value of  $E(X)$ . 2

| $x$        | 0 | 1 | 2 |
|------------|---|---|---|
| $P(X = x)$ |   |   |   |

.....

.....

.....

.....

.....

.....

.....

- (ii) Find  $E(X^2)$  and hence find  $\text{Var}(X)$  and standard deviation  $\sigma$ . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**End of question 14**

**Question 15 (16 marks)**

a) The velocity  $v$  of a particle in metres per second is given by the formula

$$v = 5(1 + e^{-t}), \text{ where } t \text{ is the time in seconds.}$$

- (i) Find the initial velocity of the particle. 1

.....

.....

.....

.....

.....

- (ii) Is the particle ever stationary? Justify your answer. 1

.....

.....

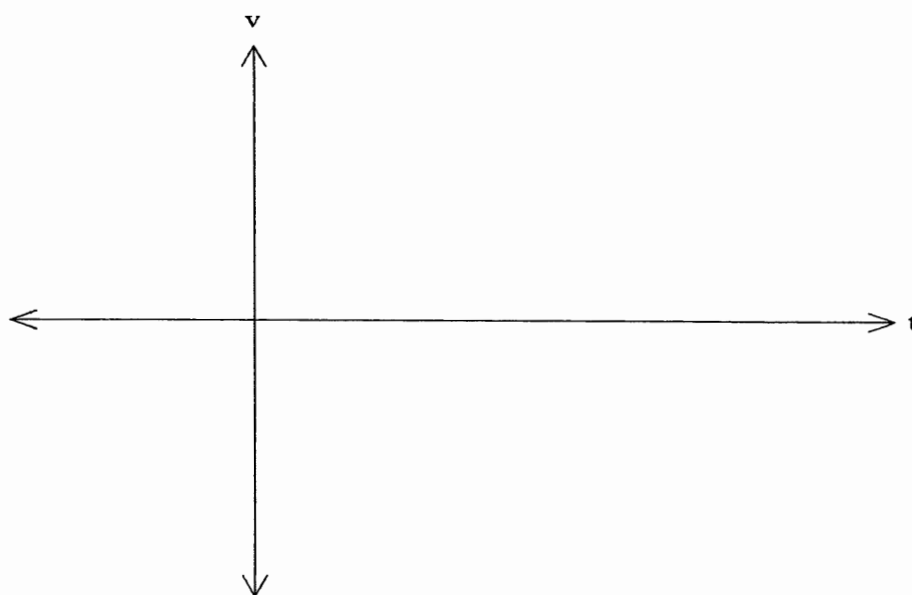
.....

.....

.....

.....

- (iii) Sketch the graph of the velocity. 2



**Question 15 continued on the next page**

(iv) Find the total distance travelled by the particle in the first 5 seconds. 2

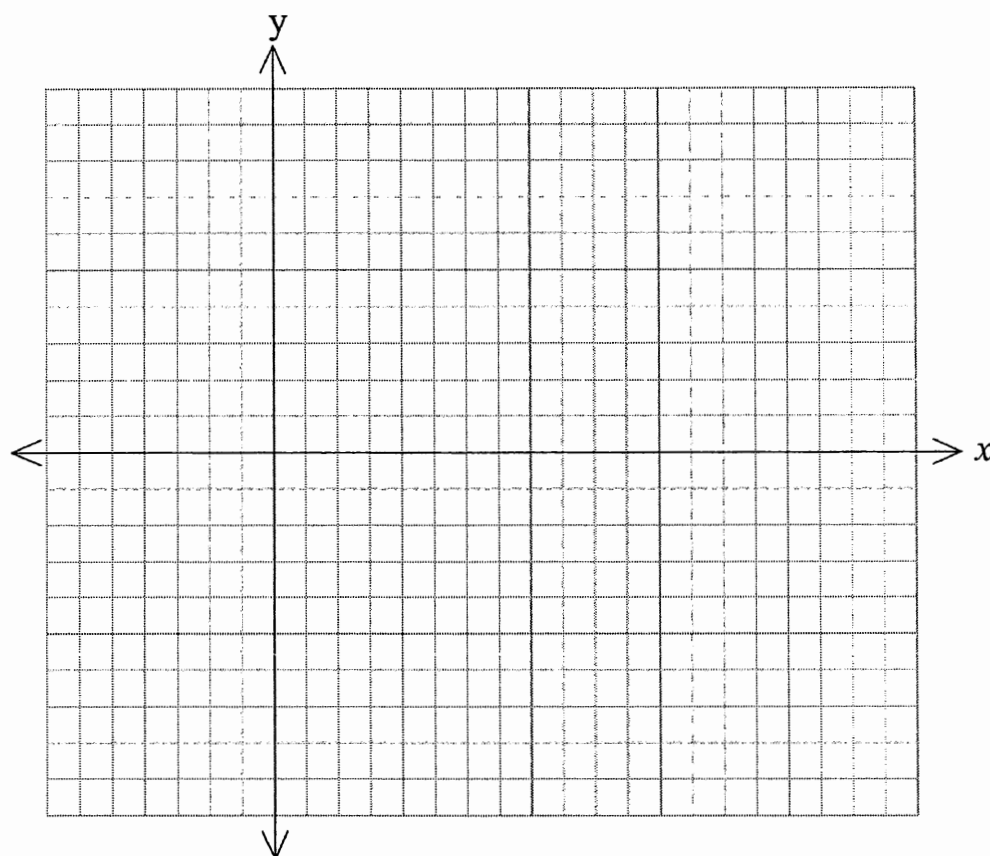
This image shows a full page of primary-ruled paper. It features approximately 20 horizontal dotted lines spaced evenly down the page, providing a guide for handwriting practice. The paper is otherwise blank, with no margins, text, or other markings.

**Question 15 continued on the next page**

b) The line  $y = mx$  is a tangent to the curve  $y = \ln(2x - 1)$  at a point  $P$ .

- (i) Sketch the line and the curve on the same diagram, clearly indicating the point  $P$ .

2



**Question 15 continued on the next page**

- (ii) Show that the coordinates of  $P$  are  $\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$ . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (iii) Hence show that  $2+m = \ln\left(\frac{4}{m^2}\right)$ . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 15 continued on the next page**

c) Given the probability density function

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function  $F(x)$ . 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

(ii) Hence find the median. 2

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**End of question 15**

**Question 16 (14 marks)**

- a) Michelle borrows \$450 000 to be repaid by regular monthly repayments of \$ $M$  over a period of 25 years at 6% per annum reducible monthly. Interest is calculated and charged just before each repayment.

Let  $A_n$  be the amount owing after  $n$  –repayments.

- (i) Show that the expression for the amount owing after two repayments is 1

$$A_2 = 450\,000(1.005)^2 - M(1.005) - M.$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

- (ii) Show that the amount owing after  $n$  –repayments is 2

$$A_n = 450\,000(1.005)^n - M \frac{(1.005)^n - 1}{0.005}.$$

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

**Question 16 continued on the next page**

(iii) Calculate the amount of each regular monthly repayment.

2

This image shows a full page of a handwriting practice worksheet. It consists of approximately 20 horizontal rows. Each row is defined by two parallel dotted lines, creating a series of uniform gaps for letter height. The lines are evenly spaced and extend across the entire width of the page. There is no text or other markings on the paper.

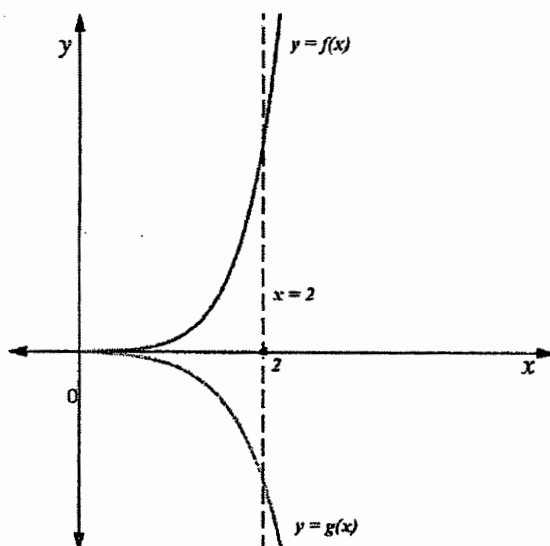
**Question 16 continued on the next page**



- [illegible]

32

- c) The graph of  $f(x) = x^2 e^{kx}$  and  $g(x) = -\frac{2xe^{kx}}{k}$  and the line  $x = 2$  is drawn below, where  $k$  is a positive constant.  $f(x) = g(x)$  at only one point, that is at  $(0, 0)$ .



Let  $A$  be the area of the region bounded by the curve  $y = f(x)$ ,  $y = g(x)$  and the line  $x = 2$ .

- (i) Write down a definite integral that gives the value of  $A$ .

1

.....

.....

.....

.....

.....

.....

- (ii) The function  $f(x)$  from part (i) is given by  $f(x) = x^2 e^{kx}$ , where  $k$  is a positive constant. Show that  $f'(x) = xe^{kx}(kx + 2)$ .

1

.....

.....

.....

.....

.....

**Question 16 continued on the next page**

(iii) Using the results of part (i) and (ii), or otherwise, find the value of  $k$  such that 2

$$A = \frac{16}{k}.$$

This image shows a single page of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

**End of Exam**

Answers - Multiple Choice (2020)  
Yr. 12 TRIAL - Maths Advanced

(1)  $\int \tan^2 x \, dx = \int \sec^2 x - 1 \, dx$  (B)  
 $= \tan x - x + C$

(2)  $\frac{d}{dx} \log_e \frac{4x^2-9}{2x-3} = \frac{d}{dx} \log_e \frac{(2x-3)(2x+3)}{(2x-3)}$  (B)  
 $= \frac{d}{dx} \log_e (2x+3) = \frac{2}{2x+3}$

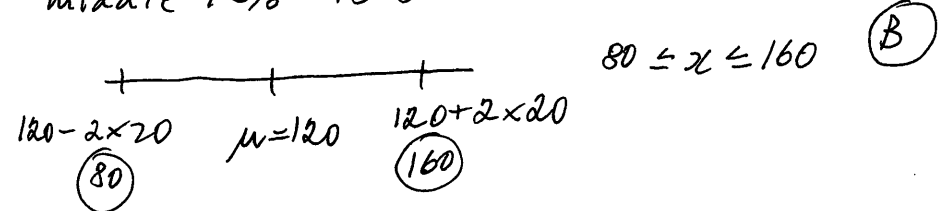
(3)  $f'(x) = \frac{x}{e^{x^2-8}} \therefore f(x) = \int \frac{x}{e^{x^2-8}} \, dx$   
 $f(x) = \int x \cdot (e^{x^2-8})^{-1} \, dx = \int x e^{8-x^2} \, dx$   
 $= -\frac{1}{2} \int \underbrace{2x \cdot e^{8-x^2}}_{g'(x)} \underbrace{1}_{g(x)} \, dx = -\frac{1}{2} e^{8-x^2} + C$  (D)  
but C can be any constant  $\therefore$

(4) curve is decreasing  $\therefore \frac{dy}{dx} < 0$   
curve is concave up  $\therefore \frac{d^2y}{dx^2} > 0$  } (D)

(5)  $\mu = \bar{x} = 55$   $(+3\sigma)$   $z=3$   $\therefore \text{score} = 55 + 3 \times 6 = 73$  (B)

(6) Area = 34 =  $\left| \int_1^2 k(x-2)^3 \, dx \right| + \int_2^4 k(x-2)^3 \, dx$   
 $34 = k \left| \int_1^2 (x-2)^3 \, dx \right| + k \int_2^4 (x-2)^3 \, dx$   
 $34 = k \left| \left[ \frac{(x-2)^4}{4} \right]_1^2 \right| + k \left[ \frac{(x-2)^4}{4} \right]_2^4$   
 $34 = k \left| 0 - \frac{1}{4} \right| + k \left[ \frac{16}{4} - 0 \right]$   
 $34 = k \left( \frac{1}{4} + 4 \right)$  (B)  
 $\therefore k = 8$

(7)  $\mu = 120$   $\sigma = 20$  ( $\text{Var}(T) = \sigma^2$ )  
middle 95% is  $2\sigma$  around  $\mu$ .



(8)  $I \doteq \frac{h}{2} (\text{first value} + \text{last value})$   
 $= \frac{1}{2} \left( \frac{\ln 1}{2} + \frac{\ln 2}{2} \right) = \frac{1}{4} \ln 2 = 0.17328\dots$  (smaller)

Exact  $I = \frac{1}{2} (\ln 2)^2 = 0.2402265$

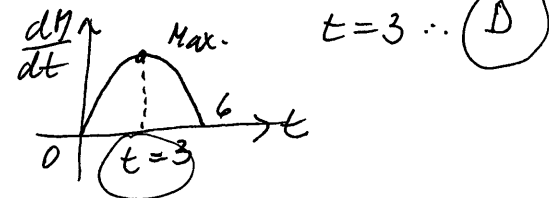
$\therefore \% = \frac{\text{approx. } I - \text{Exact } I}{\text{Exact } I} = \frac{\frac{1}{4} \ln 2 - \frac{1}{2} (\ln 2)^2}{\frac{1}{2} (\ln 2)^2} \times 100$

$\% = 27.865\% \doteq 28\%$  smaller (A)

(9)  $f(-x) = \frac{-x}{(-x)^2-5} = \frac{-x}{x^2-5} = -f(x) \therefore \text{odd}$   
by horiz. line test  $\therefore$  many to one } (D)

(10)  $M = 9t^2 - t^3$

$\frac{dM}{dt} = 18t - 3t^2 = 3t(6-t)$



## Section II- Extended Response

Attempt Questions 11-16.

Allow about 75 minutes for this section.

### Question 11(14 Marks)

a) Differentiate the following

(i)  $y = (4x - 5)(4x + 5) = 16x^2 - 25$

$$\frac{dy}{dx} = 32x$$

Marks 1

1 - correct soln

OR  $\frac{dy}{dx} = 4(4x+5) + (4x-5) \times 4$   
 $= 16x + 20 + 16x - 20 = 32x$

(ii)  $y = \sin^2 x$

2

$$\frac{dy}{dx} = 2 \sin x \cdot \cos x$$

2 - correct soln.

1 - correctly diff.

$\sin x$

1 -  $\frac{dy}{dx} = 2 \sin x \cos x$

b) In AP,  $T_3 = 5$  and  $T_{10} = 26$ .  
Find the sum of  $S_{14}$ .

2

AP:  $T_3 = 5$   $T_{10} = 26$

$$T_3 = 5 = a + 2d$$

2 - correct soln.

$$T_{10} = 26 = a + 9d$$

1 - correctly finds

$$21 = 7d \therefore d = 3, a = -1$$

a or d

$$\therefore S_{14} = \frac{14}{2} (2a + (14-1)d) = 259$$

1 - applies  $S_n$  for A.P correctly

Question 11 continued on the next page

c) Evaluate

$$\int_1^4 5(9x - 4)^4 dx$$

$$= \left[ 5 \times \frac{(9x-4)^5}{5 \times 9} \right]_1^4$$

2 - correct soln.

1 - correct integral

$$= \frac{1}{9} \left[ (9x-4)^5 \right]_1^4$$

1 - correct answer

from incorrect integral

$$= \frac{1}{9} \left[ (9(4)-4)^5 - (9-4)^5 \right]$$

$$= \frac{1}{9} \left[ 32^5 - 5^5 \right] = 3727923$$

d)  $e^{2x} + 3e^x - 10 = 0$

2 - correct soln.

1 - correctly

let  $m = e^x$

reduces to quadratic eqn

$$m^2 + 3m - 10 = 0$$

$$(m+5)(m-2) = 0$$

solves it

$$m = -5$$

$$m = +2$$

correctly

$$e^x = -5$$

$$e^x = 2$$

no solns.

$$x = \ln 2$$

$\therefore$  solution  $x = \ln 2$  (only)

Question 11 continued on the next page

e)

(i) Show that  $\frac{d}{dx}(\sec^2 x) = 2 \tan x \sec^2 x$ 

2

$$\begin{aligned}
 \text{LHS} &= \frac{d}{dx}(\sec^2 x) = \frac{d}{dx}\left(\frac{1}{\cos^2 x}\right) \quad \begin{array}{l} 2 - \text{correct soln.} \\ 1 - \text{differentiates} \\ \text{correctly } \sec^2 x \end{array} \\
 &= \frac{d}{dx}(\cos^{-2} x) = -2 \cos^{-3} x \cdot (-\sin x) \\
 &= \frac{2 \sin x}{\cos^3 x} = \frac{2 \sin x}{\cos x} \times \frac{1}{\cos^2 x} \quad \begin{array}{l} 1 - \text{applies trig.} \\ \text{identities correctly} \end{array} \\
 &= 2 \tan x \cdot \sec^2 x \\
 &= \text{RHS} \therefore \text{shown}
 \end{aligned}$$

(ii) Hence find  $\int \tan x \sec^2 x dx$ 

1

$$\begin{aligned}
 \text{from (i)} \quad \frac{d}{dx}(\sec^2 x) &= 2 \tan x \sec^2 x \quad 1 - \text{correct soln.} \\
 \therefore \sec^2 x &= \int 2 \tan x \sec^2 x dx \quad - \text{ignore } +C \\
 \therefore \frac{1}{2} \sec^2 x &= \int \tan x \sec^2 x dx \\
 \therefore \int \tan x \sec^2 x dx &= \frac{1}{2} \sec^2 x + C \\
 \text{OR } &= \frac{1}{2} (1 + \tan^2 x) + C
 \end{aligned}$$

Question 11 continued on the next page

f) Given a function  $f(x) = \begin{cases} 6x - 6x^2 & 0 \leq x \leq 1 \\ 0 & \text{Otherwise} \end{cases}$ (i) Show that  $f(x)$  represents probability density function.

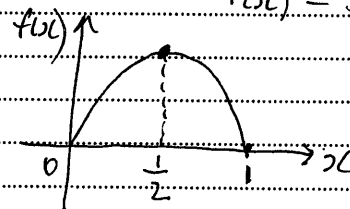
2

$$\begin{aligned}
 f(x) \text{ represents PDF if } & \quad 2 - \text{correct soln.} \\
 & \bullet \int_0^1 f(x) dx = 1 \text{ and } f(x) \geq 0 \quad \begin{array}{l} 1 - \text{finds } \int f(x) dx \\ \text{on domain} \end{array} \\
 \therefore \int_0^1 (6x - 6x^2) dx & \quad \text{true} \quad \text{correctly} \\
 &= \left[ 3x^2 - 2x^3 \right]_0^1 \\
 &= [(3-2) - 0] = 1 \therefore \text{Yes it's PDF}
 \end{aligned}$$

(ii) Find the mode of the probability density function  $f(x)$ 

1

$$\begin{aligned}
 \text{By sketching } f(x) &= 6x - 6x^2 \\
 f(x) &= 6x(1-x)
 \end{aligned}$$



Mode is  $x = \frac{1}{2}$  with the highest value of  $f(x)$

1 - correct soln.

End of Question 11

Question 12 (3 Marks)

- a) Find the value(s) of  $m$  such that  $y = 2x + m$  is a tangent to the parabola

$$y = 2x^2 + 6x - 5.$$

(2)

2 + correct soln.

$y = 2x + m$   
 $2 \Rightarrow$  gradient of tangent

1 - finds point of contact by using calculus

$\therefore y' = 4x + 6$  where  $m = 2$   
 $\therefore 2 = 4x + 6$   
 $-1 = x \quad \therefore y = 2(-1)^2 + 6(-1) - 5$   
 $y = -9$   
 $\therefore$  pt. of contact  $(-1, -9)$

1 - finds  $\Delta$  correctly by using simult. eqns.

1 - finds gradient function correctly &  $x$ -coord. of pt. of contact

Sub.  $(-1, -9)$  into  $y = 2x + m$   
 $-9 = 2(-1) + m$   
 $\therefore m = -7$

OR by simultaneous eqn.

$2x + m = 2x^2 + 6x - 5$

1 - creates quad. eqn. correctly by solving simult. eqns, & attempts to solve  $\Delta = 0$

$2x^2 + 4x - 5 - m = 0$   
 $\Delta = 0$  (since tangent, only one solution)  
 $0 = b^2 - 4ac$   
 $0 = 4^2 - 4(2)(-5 - m)$   
 $0 = 16 + 40 + 8m$   
 $m = -7$

Question 12 continued on the next page

- b) Angela guesses three questions in her multiple choice test, which has four options per question. Find the probability that Angela gets

1/4 C 1/4 C 1/4 C  
 1/4 Correct (C) 3/4 IC 3/4 IC  
 3/4 incorrect (IC) 1/4 C 1/4 C  
 3/4 IC 3/4 IC  
 3/4 IC

- i) Only one correct

$$P = P(Ci\bar{i}) + P(\bar{i}Ci) + P(i\bar{i}C)$$

$$= \left(\frac{1}{4}\right)^1 \times \left(\frac{3}{4}\right)^2 + \left(\frac{1}{4}\right)^1 \times \left(\frac{3}{4}\right)^2 + \left(\frac{3}{4}\right)^2 \times \left(\frac{1}{4}\right)^1$$

$$= 3 \times \left(\frac{1}{4}\right) \times \left(\frac{3}{4}\right)^2 = \frac{27}{64}$$

1 - correct soln.

- (ii) At least one correct

$$P(\text{at least one correct})$$

$$= 1 - P(\text{no correct})$$

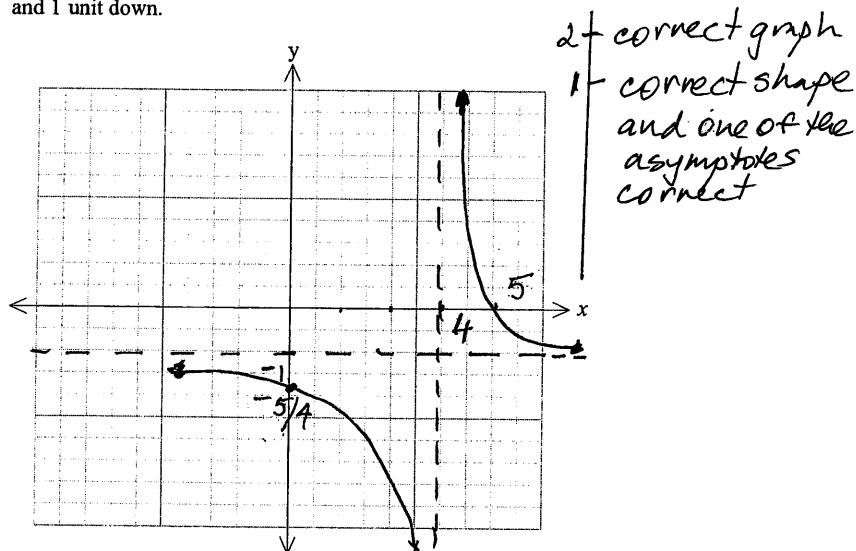
$$= 1 - \left(\frac{3}{4}\right)^3 = \frac{37}{64}$$

1 - correct soln.

Question 12 continued on the next page

c)

- (i) Sketch the hyperbola by shifting  $y = \frac{1}{x-1}$  horizontally 3 units to the right and 1 unit down. 2



- (ii) State the equation of the shifted hyperbola, then find all the intercepts of the shifted hyperbola with the axes and mark them on your graph in part (i). 2

$y = \frac{1}{x-4} - 1$  2 - correct soln.  
 1 - correct eqn.  
 $x=0 \therefore y = \frac{1}{0-4} - 1 = -\frac{1}{4} \text{ or } -\frac{5}{4}$  1 - one of the intercepts correct and labeled on the graph  
 $\therefore y\text{-int } (0, -\frac{5}{4})$   
 $y=0 \therefore 0 = \frac{1}{x-4} - 1$   
 $1 = \frac{1}{x-4} \therefore x\text{-int } (5, 0)$   
 $x-4=1 \therefore x=5$

Question 12 continued on the next page

- (d) Consider the piece-wise defined function.

$$f(x) = \begin{cases} x^2 - 1 & x \leq 1 \\ 4 - x^2 & x > 1 \end{cases}$$

- (i) Find  $f(1)$

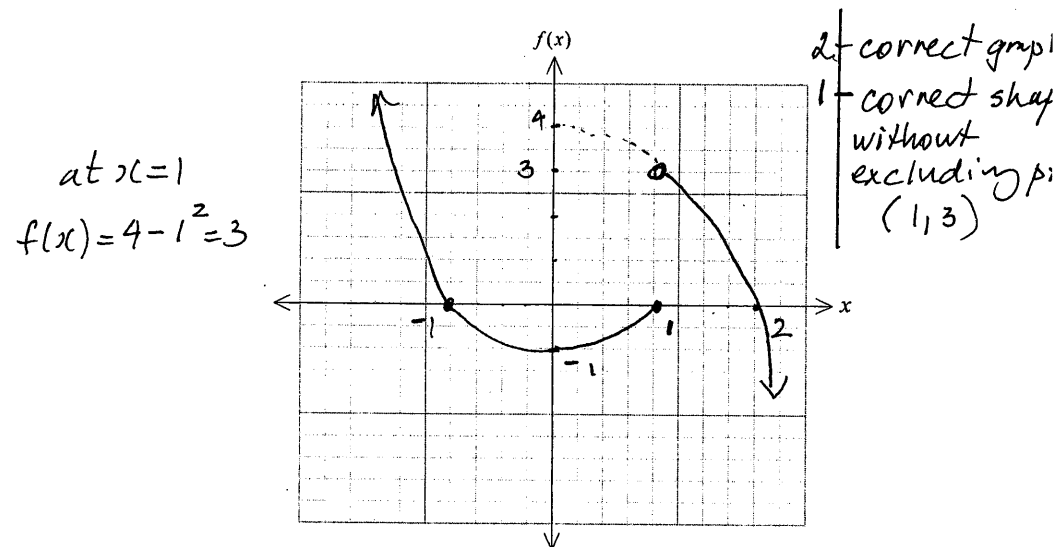
$$f(1) = 1^2 - 1 = 0$$

1 - correct soln.

- (ii) Find  $x$  if  $f(x) = 0$

$f(x) = 0 \begin{cases} x^2 - 1 = 0 \therefore x = \pm 1 \\ 4 - x^2 = 0 \therefore x = \pm 2 \end{cases}$  2 - correct answers  
 1 - all 4 solns.  
 but  $x > 1 \therefore x = 2$  correct with one excluding  $x = -1$   
 $\therefore x = \pm 1$

- (iii) Sketch the function showing all intercepts. 2

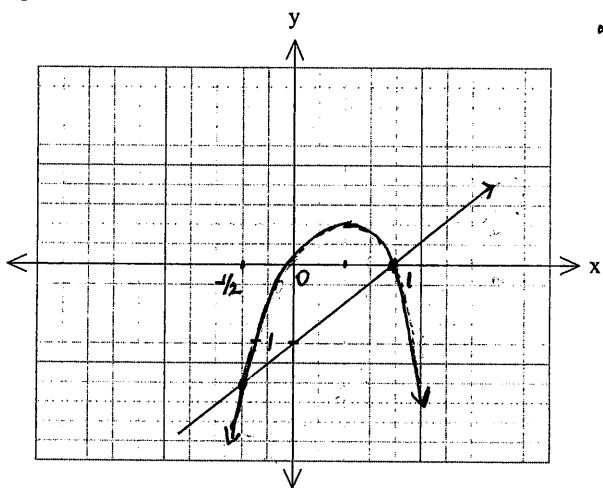


End of Question 12

**Question 13 (18 Marks)**

- a) (i) Sketch the graphs of  $f(x) = 2x - 2x^2$  and  $g(x) = x - 1$  on the same number plane.

$$f(x) = 2x(1-x)$$



2 - both graphs correct  
1 - correct graph of  $g(x)$  with correct intercept  
1 -  $f(x)$  correct graph with correct x-intercepts

- (ii) Using your graphs from part (i), or otherwise solve the inequality

$$x - 1 < 2x - 2x^2$$

$f(x) \cap g(x) = \text{pts. of intersection}$

$$x - 1 = 2x - 2x^2$$

$$2x^2 - x + 1 = 0$$

$$(2x+1)(x-1) = 0$$

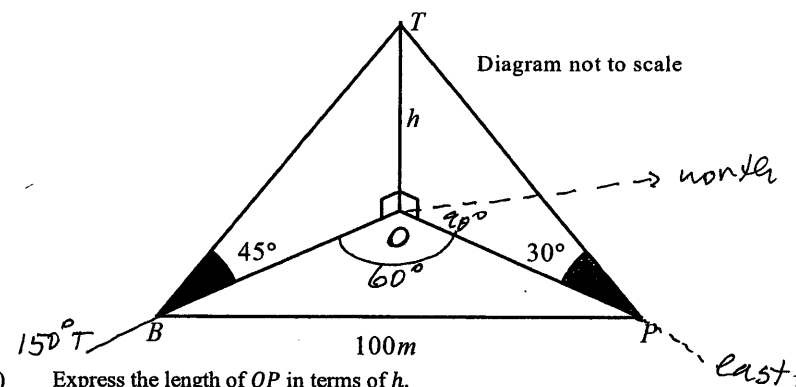
$$x = -\frac{1}{2} \quad x = 1$$

$\therefore$  answer graphically (line below parabola)  
 $-\frac{1}{2} < x < 1$

2 - correct solus.  
1 - finds pts. of intersection  
 $f(x) \cap g(x)$  correctly

Question 13 continued on the next page

- b) A surveyor stands at a point  $P$ , which is due east of the tower  $OT$ , of height  $h$  metres. The angle of elevation of the top of the tower  $T$  from  $P$  is  $30^\circ$ . The surveyor then walks 100 metres to point  $B$ , which is on a bearing of  $150^\circ$  from the foot of tower  $O$ . The angle of elevation of the top of the tower from  $B$  is now  $45^\circ$ .



- (i) Express the length of  $OP$  in terms of  $h$ .

from  $\triangle OPT$

$$\tan 30^\circ = \frac{h}{OP}$$

$$OP = \frac{h}{\tan 30^\circ}$$

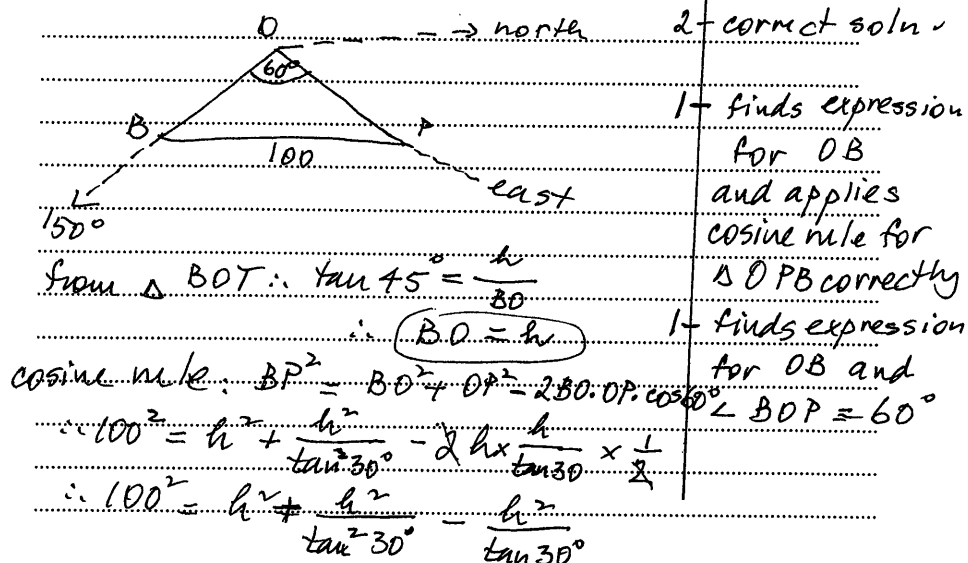
$$\text{OR } OP = \sqrt{3}h$$

1 - correct expression with  $\tan 30^\circ$   
1 - correct answer  $OP = \sqrt{3}h$

Question 13 continued on the next page

(ii) Show that  $(100)^2 = h^2 + \frac{h^2}{\tan^2 30^\circ} - \frac{h^2}{\tan 30^\circ}$

2



(iii) Hence find the height of the tower. Answer correct to 1 decimal place.

1

$$100^2 = h^2 \left( 1 + \frac{1}{\tan^2 30^\circ} - \frac{1}{\tan 30^\circ} \right)$$

$$h^2 = \frac{100^2}{\left( 1 + \frac{1}{\tan^2 30^\circ} - \frac{1}{\tan 30^\circ} \right)}$$

$$h^2 = 4409.269852$$

$$\therefore h = 66.4 \text{ m (1dp)}$$

1 - correct answer  
(ignore rounding)

c) The following information shows a group of people's waist measurements and weights.

|               |    |    |    |    |    |    |    |     |
|---------------|----|----|----|----|----|----|----|-----|
| Waist (cm) x  | 72 | 67 | 85 | 96 | 80 | 90 | 98 | 105 |
| Weight (kg) y | 58 | 50 | 72 | 85 | 70 | 79 | 82 | 84  |

(i) Calculate the correlation coefficient, r, for their waist and weight measurements and hence describe the strength of the relationship.

2

$r = 0.9592$   
(calculator)  
 $\therefore$  strong correlation positive  
2 - correct soln.  
1 - correct r  
1 - from their "r" correct conclusion for strength of the relationship

(ii) Find the equation of the Least-Squares Regression Line.

1

from calculator

$$A = -8.2368$$

$$B = 0.93203$$

$$y = A + Bx$$

$$y = -8.2368 + 0.93203x$$

1 - correct solns.

Question 13 continued on the next page

d) Given the function  $f(x) = \ln(x^2 + 1)$ .

(i) Find the domain of  $f(x)$ .

1

$$x^2 + 1 > 0$$

which is always

$\therefore$  Domain = all real  $x$

1- correct solns.

(ii) Find any stationary point(s) and determine their nature.

2

$$f'(x) = \frac{2x}{x^2 + 1}$$

2- correct solns.

$$0 = \frac{2x}{x^2 + 1}$$

1- finds stationary point correctly

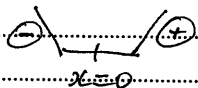
$$2x = 0, \quad y = \ln(0+1) = 0$$

1- determines the nature of st. point correctly

$\therefore (0, 0)$  is stationary point

Nature by  $f''$  or table

|       |    |   |   |
|-------|----|---|---|
| $x$   | -1 | 0 | 1 |
| $f''$ | -1 | 0 | 1 |



$\therefore (0, 0)$  is minimum turning pt.

(or)  $f''(x) = \frac{2 - 2x^2}{(x^2 + 1)^2}$

$$f''(0) = 2 > 0 \quad \therefore (0, 0) \text{ is min. t.p.}$$

Question 13 continued on the next page

(iii) Find any point(s) of inflection.

2

$$f''(x) = \frac{2(x^2 + 1) - 2x(2x)}{(x^2 + 1)^2}$$

2- correct solns.

$$f''(x) = \frac{2 - 2x^2}{(x^2 + 1)^2} = 0$$

1- solves for  $f'' = 0$  correctly

$$2 - 2x^2 = 0 \quad \therefore x = \pm 1$$

1- justifies pt. of inflexion correctly

$\therefore (1, \ln 2), (-1, \ln 2)$  possible pts. of inflexion

|       |                 |    |   |   |                 |
|-------|-----------------|----|---|---|-----------------|
| $x$   | -2              | -1 | 0 | 1 | 2               |
| $f''$ | $-\frac{6}{25}$ | 0  | 2 | 0 | $-\frac{6}{25}$ |

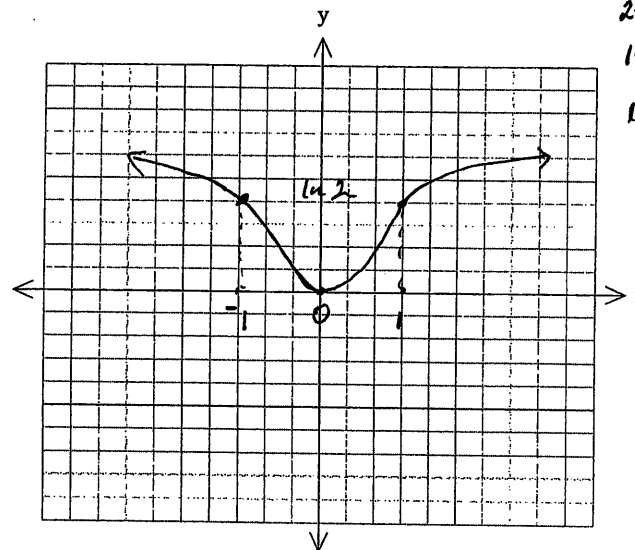
concavity changes at  $x = \pm 1$

$\therefore (1, \ln 2), (-1, \ln 2)$  are points of inflexion

(iv) Sketch the graph of  $f(x) = \ln(x^2 + 1)$  showing all features from

2

part (ii) and (iii).



2- correct graph showing all features  
1- correct shape  
1- shows correctly features from part (ii) & (iii)

End of Question 13

Question 14 (14 marks)

- a) (i) Prove the following identity

$$(1 + \tan x)^2 = 2 \tan x + \sec^2 x$$

$$\begin{aligned} LHS &= (1 + \tan x)^2 \\ &= 1 + 2 \tan x + \tan^2 x \\ &= \underbrace{1 + \tan^2 x} + 2 \tan x \\ &= \sec^2 x + 2 \tan x \\ &= RHS \end{aligned}$$

1 - correct solns

- (ii) Hence find the area bounded by  $y = (1 + \tan x)^2$  and the x-axis between  $-\frac{\pi}{4} \leq x \leq \frac{\pi}{4}$

$$\begin{aligned} \text{Area} &= \int_{-\pi/4}^{\pi/4} (1 + \tan x)^2 dx \\ &= \int_{-\pi/4}^{\pi/4} 2 \tan x + \sec^2 x dx \\ &= \left[ -2 \ln |\cos x| + \tan x \right]_{-\pi/4}^{\pi/4} \\ &= \left[ (-2 \ln |\cos \frac{\pi}{4}| + 1) - (-2 \ln |\cos \frac{\pi}{4}| + (-1)) \right] \\ &= 2 \end{aligned}$$

2 - correct solns.  
1 - correctly integrates  
tan x  
1 - correctly uses  
part (i)  
1 - correctly evaluates  
the definite integral

Question 14 continued on the next page

- b) given  $y = 2 \sin \left( 2x - \frac{\pi}{3} \right)$

- (i) State the amplitude and period.

2

$$\text{amplitude} = 2$$

2 - correct solns.

$$\text{period } T = \frac{2\pi}{2} = \pi$$

1 - correct amplitude

1 - correct period

- (ii) Find all intercepts of  $y = 2 \sin \left( 2x - \frac{\pi}{3} \right)$  with the axes for  $0 \leq x \leq \pi$

2

$$\begin{aligned} y\text{-int: } x &= 0 \\ y &= 2 \sin \left( 2(0) - \frac{\pi}{3} \right) \\ y &= -2 \frac{\sqrt{3}}{2} = -\sqrt{3} \\ \therefore (0, -\sqrt{3}) \text{ y-int.} \end{aligned}$$

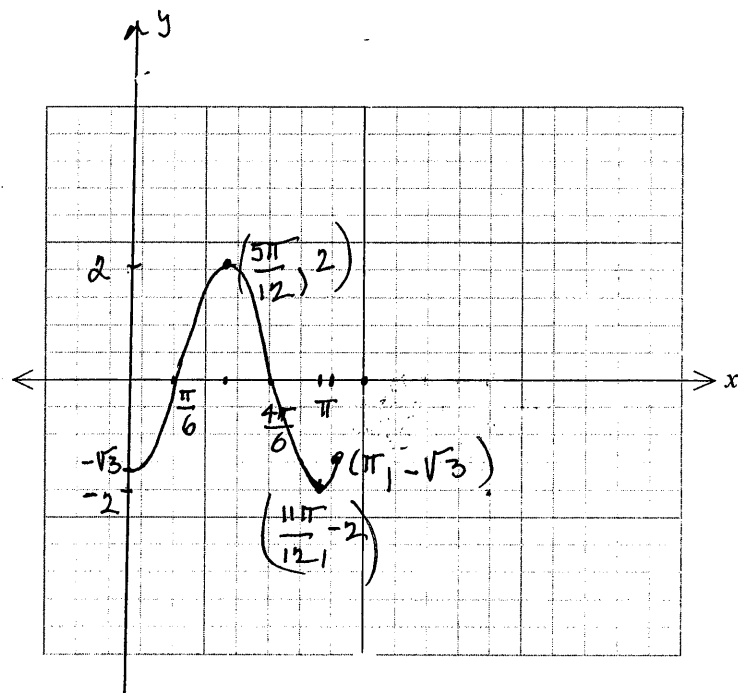
2 - correct solns.

1 - correct y-int.

$$\begin{aligned} x\text{-int: } y &= 0 \\ 0 &= 2 \sin \left( 2x - \frac{\pi}{3} \right) \\ 2x - \frac{\pi}{3} &= 0, \pi, 2\pi \\ \therefore x &= \frac{\pi}{6}, \frac{4\pi}{6}, \frac{7\pi}{6} \rightarrow \text{out of domain} \\ \therefore \left( \frac{\pi}{6}, 0 \right), \left( \frac{2\pi}{3}, 0 \right) \text{ are x-int.} \end{aligned}$$

Question 14 continued on the next page

- (iii) Hence sketch the graph of  $y = 2\sin\left(2x - \frac{\pi}{3}\right)$  for  $0 \leq x \leq \pi$  showing all features from part (i) and (ii) and global maximum and minimum. 2



- 2 - correct graph  
1 - correct shape and x, y-intercepts  
1 - showing correct Max/Min

- c) A bag contains three red balls and four black balls. Two balls are selected at random without replacement from the bag.

Let  $X$  be the number of black balls drawn. exact value of

- (i) Fill in the following table and hence find  $E(X)$ . 2

| $x$        | 0        | 1              | 2        |
|------------|----------|----------------|----------|
| $P(X = x)$ | RR $1/7$ | RB or BR $4/7$ | BB $2/7$ |

Handwritten calculations for probabilities and expected value:

$$P(0) = \frac{3}{7} \times \frac{2}{6} = \frac{1}{7}$$

$$P(1) = 2 \times \frac{3}{7} \times \frac{4}{6} = \frac{4}{7}$$

$$P(2) = \frac{4}{7} \times \frac{3}{6} = \frac{2}{7}$$

2 - correct soln.  
1 - Finds correct  $E(X) = \mu$   
from incorrect table of values

$$E(X) = \sum x p(x) = 0 \times \frac{1}{7} + 1 \times \frac{4}{7} + 2 \times \frac{2}{7} = \frac{8}{7}$$

- (ii) Find  $E(X^2)$  and hence find  $\text{Var}(X)$  and  $\sigma$  2

$$E(X^2) = \sum x^2 p(x) = 0 \times \frac{1}{7} + 1^2 \times \frac{4}{7} + 2^2 \times \frac{2}{7}$$

$$\therefore E(X^2) = \frac{12}{7}$$

$$\text{Var}(X) = E(X^2) - \mu^2$$

$$= \frac{12}{7} - \left(\frac{8}{7}\right)^2 = \frac{20}{49}$$

$$\text{Var}(X) = \frac{20}{49} \therefore \sigma = 0.638976\dots$$

$$\sigma = \sqrt{\frac{20}{49}} = \frac{\sqrt{20}}{7} = \frac{2\sqrt{5}}{7} \therefore \sigma = 0.64396$$

End of question 14

Question 14 continued on the next page

**Question 15 (16 marks)**

a) The velocity  $v$  of a particle in metres/seconds is given by the formula

$$v = 5(1 + e^{-t}), \text{ where } t \text{ is time in seconds.}$$

(i) Find the initial velocity of the particle.

1

$$t = 0$$

$$v = 5(1 + e^{-0}) = 10 \text{ m/s}$$

1 - correct soln.

(ii) Is the particle ever stationary? Justify your answer.

1

$$v = 0$$

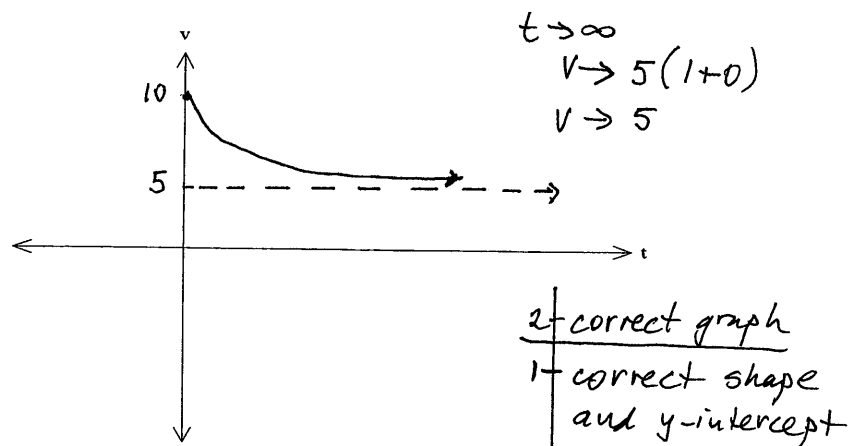
$$0 = 5(1 + e^{-t})$$

$$\therefore \text{no soln.} \therefore v \neq 0$$

1 - correct soln.

(iii) Sketch the graph of the velocity.

2



Question 15 continued on the next page

(iv) Find the total distance travelled by the particle in the first 5 seconds.

2

$$d = \int_0^5 5(1 + e^{-t}) dt$$

2 - correct solns.  
1 - correct integration

$$= 5 \left[ t - e^{-t} \right]_0^5$$

$$= 5 \left[ 5 - e^{-5} - (0 - e^0) \right]$$

$$= 5(5 - e^{-5} + 1)$$

$$= 30 - 5e^{-5} \text{ (metres) exact}$$

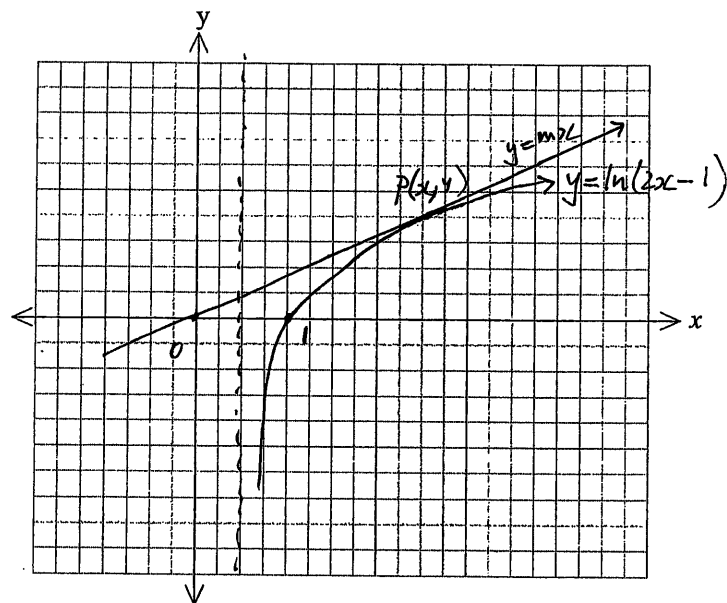
(OR)  $d = 29.97 \text{ (2d.p.)}$

Question 15 continued on the next page

b) The line  $y = mx$  is a tangent to the curve  $y = \ln(2x - 1)$  at a point  $P$ .

(i) Sketch the line and the curve on the same diagram, clearly indicating the point  $P$ .

2



$$y = \ln(2x - 1)$$

$$2x - 1 > 0$$

$$x > \frac{1}{2}$$

$$2x - 1 = 1$$

$$x = 1$$

$$(x\text{-int})$$

$$y = mx \rightarrow \text{passing through } (0, 0)$$

2- correct graphs  
1- correct log. graph  
1- correct graph of the line passing through (0,0) & clearly indicating point of contact  $P(x, y)$ .

Question 15 continued on the next page

(ii) Show that the coordinates of  $P$  are  $\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$ .

2

2- correct solns.  
1- equates  $y' = m$  and solves for  $x$   
1- substitutes  $x$ -value into one of the eqns.

$$y = \ln(2x - 1)$$

$$\therefore y' = \frac{2}{2x - 1}$$

if  $y = mx$  is a tangent to  $y = \ln(2x - 1)$  at point  $P(x, y)$

$$\therefore m = \frac{2}{2x - 1}$$

$$2x - 1 = \frac{2}{m}$$

$$2x = \frac{2}{m} + 1 \therefore x_p = \frac{1}{m} + \frac{1}{2} = \frac{2+m}{2m}$$

sub.  $x_{\text{coord}}$  into  $y = mx$

$$\therefore y_p = mx \cdot \frac{2+m}{2m} = \frac{2+m}{2}$$

(iii) Hence show that  $2 + m = \ln\left(\frac{4}{m^2}\right)$ .

2

2- correct solns.  
1- sub. coordinates of  $P$  into  $y = \ln(2x - 1)$  and attempts to solve it

since  $P\left(\frac{2+m}{2m}, \frac{2+m}{2}\right)$  (part ii)

and  $P$  lies on  $y = \ln(2x - 1)$

$\therefore$  coord. of  $P$  satisfy equation

sub. in  $y = \ln(2x - 1)$

$$\frac{2+m}{2} = \ln\left(2\left(\frac{2+m}{2m}\right) - 1\right)$$

$$\frac{2+m}{2} = \ln\left(\frac{2+m}{m} - 1\right) = \ln\left(\frac{2}{m} + \frac{m}{m} - 1\right)$$

$$\frac{2+m}{2} = \ln\left(\frac{2}{m} + 1 - 1\right) = \ln\left(\frac{2}{m}\right)$$

$$\therefore 2+m = 2 \ln\left(\frac{2}{m}\right) = \ln\left(\frac{2}{m}\right)^2 = \ln\frac{4}{m^2}$$

Question 15 continued on the next page

$$\therefore 2+m = \ln\left(\frac{4}{m^2}\right) \therefore \text{shown}$$

c) Given the probability density function

$$f(x) = \begin{cases} 2e^{-2x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

(i) Find the cumulative distribution function  $F(x)$ .

2

$$F(x) = \int_0^x 2e^{-2x} dx$$

$$F(x) = 2x - \frac{1}{2} [e^{-2x}]_0^x$$

$$F(x) = -[e^{-2x} - e^0]$$

$$\therefore F(x) = -e^{-2x} + 1$$

2 - correct solns.  
1 - correctly integrates  $f(x)$

(ii) Hence find the median.

2

let  $m$  be median

$$\therefore F(m) = \frac{1}{2}$$

$$\frac{1}{2} = -e^{-2x} + 1$$

$$e^{-2x} = 1 - \frac{1}{2}$$

$$\ln(e^{-2x}) = \ln \frac{1}{2}$$

$$-2x = \ln \frac{1}{2}$$

$$x = -\frac{1}{2} \ln \frac{1}{2} \quad (\text{median})$$

$$\therefore \text{median} = -\frac{1}{2} \ln \frac{1}{2} \text{ or } \ln \sqrt{2} \text{ or } 0.347$$

$$\text{OR } -\frac{1}{2} (-\ln 2) = \frac{1}{2} \ln 2$$

2 - correct solns.  
1 - equates  $F(x) = \frac{1}{2}$   
and shows significant progress to find the value of the median.

End of question 15

Question 16 (14 marks)

~~\$ 450 000~~

a) Michelle borrows \$50 000 to be repaid by regular monthly repayments of \$ $P$  over a period of 25 years at 6% per annum reducible monthly. Interest is calculated and charged just before each repayment.

Let  $A_n$  be the amount owing after  $n$  -repayments.

(i) Show that the expression for the amount owing after two repayments is

1

$$A_2 = 450\,000(1.005)^2 - P(1.005) - P$$

$$A_1 = 450\,000 \left(1 + \frac{6 \div 12}{100}\right) - P$$

$$= 450\,000(1.005) - P$$

$$A_2 = A_1(1.005) - P$$

$$= 450\,000 \times 1.005^2 - P(1.005) - P$$

$\therefore$  shown

1 - correct solns.

(ii) Show that the amount owing after  $n$  -repayments is

2

$$A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{0.005}$$

following pattern

$$\text{from (i)} \quad A_n = 450\,000(1.005)^n - P(1.005)^{n-1} - \dots - P$$

$$\therefore A_n = 450\,000(1.005)^n - P(1.005^{n-1} + \dots + 1)$$

G.P.  $a=1$   $r=1.005$   
 $n=n$

$$\therefore A_n = 450\,000(1.005)^n - P \frac{r^n - 1}{r - 1}$$

$$A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{1.005 - 1}$$

$$\therefore A_n = 450\,000(1.005)^n - P \frac{(1.005)^n - 1}{0.005}$$

$\therefore$  shown

2 - correct soln.  
1 - correctly applies pattern from (i)  
1 - correctly applies the sum of GP formula

Question 16 continued on the next page

(iii) Calculate the amount of each repayments  $P$ .

2

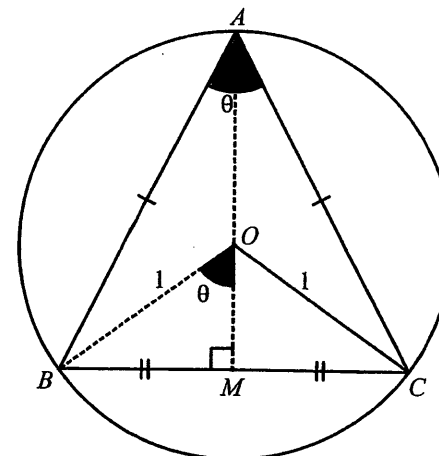
after 25 years =  $25 \times 12 = 300 = n$  2- correct soln.  
 $\therefore A_{300} = 0 = 450000(1.005)^{300} - P \frac{1.005^{300} - 1}{0.005}$  1- using part (i)  
 correctly equates  
 $A_{300} = 0$   
 and attempts  
 to solve it for  $P$   
 $P \frac{1.005^{300} - 1}{0.005} = 450000(1.005)^{300}$   
 $P = \frac{450000(1.005)^{300}}{1.005^{300} - 1} \times 0.005$

$P = \$2899.3563...$

$\therefore P = \$2899.36$

Question 16 continued on the next page

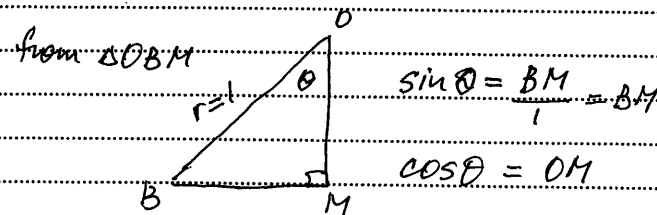
b) An isosceles triangle  $\triangle ABC$  is inscribed within a unit circle centred at  $O$ , as shown in the diagram below. Let  $M$  be the midpoint of  $BC$ ,  $\angle BAC = \theta$  and  $\angle BOM = \theta$ .



(i) Show that the area of  $\triangle ABC$  is  $A = \sin\theta(1 + \cos\theta)$ .

2

Area  $\triangle ABC = \frac{1}{2} BC \times AM$  (AM  $\perp$  BC) 2- correct soln.  
 base height 1- finds BC or AM in terms of  $\theta$



$AM = AO + OM = 1 + \cos\theta$

$BC = 2 \times BM = 2 \times \sin\theta$

$\therefore \text{Area} = \frac{1}{2} \times BC \times AM = \frac{1}{2} \times 2 \sin\theta \times (1 + \cos\theta)$   
 $\therefore A = \sin\theta(1 + \cos\theta) \therefore \text{shown}$

Question 16 continued on the next page

- (iii) Hence prove that the area of the isosceles triangle is maximum when it is equilateral.

3

3 - correct soln.  
2 - finds stationary points correctly  
1 - differentiates Area formula correctly & attempts to solve  $A' = 0$   
1 - determines the nature of st.pts and concludes for  $\Delta$  to be equilateral

$$A = \sin \theta (1 + \cos \theta)$$

$$A' = \cos \theta (1 + \cos \theta) + \sin \theta (0 - \sin \theta)$$

$$\therefore A' = \cos \theta + \cos^2 \theta - \sin^2 \theta$$

$$A' = 0$$

$$0 = \cos \theta + \cos^2 \theta - \sin^2 \theta$$

$$0 = \cos \theta + \cos^2 \theta - (1 - \cos^2 \theta)$$

$$0 = 2\cos^2 \theta + \cos \theta - 1$$

$$\theta = (2\cos^2 \theta - 1)(\cos \theta + 1)$$

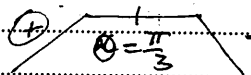
$$\cos \theta = \frac{1}{2} \quad \cos \theta = -1$$

$$\theta = \frac{\pi}{3}, \left(\frac{5\pi}{3}\right) \text{ - impossible in triangle } \theta \text{ - acute}$$

$$\theta = 180^\circ$$

$$\therefore \theta = \frac{\pi}{3} \text{ (the only solution)}$$

|        |          |       |                 |        |
|--------|----------|-------|-----------------|--------|
| Nature | $\theta$ | 1     | $\frac{\pi}{3}$ | 1.1    |
| $A'$   |          | 0.124 | 0               | -0.135 |

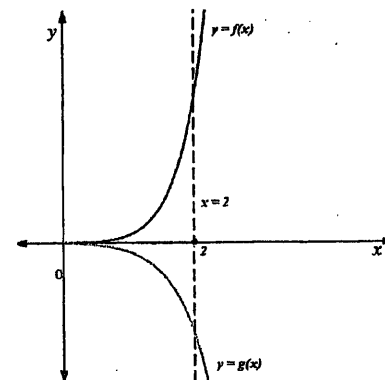


$\therefore$  at  $\theta = \frac{\pi}{3}$  Area is Maximum

but if  $\theta = \frac{\pi}{3}$   $\therefore \Delta ABC$  is equilateral/  
and given  $\Delta ABC$  is isosceles

Question 16 continued on the next page

- c) The graph of  $f(x) = x^2 e^{kx}$  and  $g(x) = -\frac{2xe^{kx}}{k}$  and the line  $x = 2$  is drawn below.  
 $f(x) = g(x)$  at only one point, that is at  $(0, 0)$ .



Let  $A$  be the area of the region bounded by the curve  $y = f(x)$ ,  $y = g(x)$  and the line  $x = 2$ .

- (i) Write down a definite integral that gives the value of  $A$ .

1

1 - correct soln.

$$A = \int_0^2 f(x) - g(x) dx$$

Q.R.

$$\therefore A = \int_0^2 x^2 e^{kx} - \left(-\frac{2xe^{kx}}{k}\right) dx$$

- (ii) The function  $f(x)$  from part (i) is given by  $f(x) = x^2 e^{kx}$  where  $k$  is a positive constant. Show that  $f'(x) = xe^{kx}(kx + 2)$

1 - correct soln.

$$f(x) = x^2 e^{kx} \quad k > 0$$

$$f'(x) = 2xe^{kx} + x^2 \cdot k e^{kx}$$

$$\therefore f'(x) = xe^{kx}(2 + kx)$$

$\therefore$  shown

Question 16 continued on the next page

(iii) Using the results of part (i) and (ii), or otherwise, find the value of  $k$  such that 2

$$A = \frac{16}{k}$$

from (i)  $A = \int_0^2 x^2 e^{kx} - \frac{2xe^{kx}}{k} dx$

$$\therefore A = \int_0^2 x^2 e^{kx} + \frac{2xe^{kx}}{k} dx$$

$$= \frac{1}{k} \int_0^2 kx^2 e^{kx} + 2xe^{kx} dx$$

$$= \frac{1}{k} \int_0^2 x e^{kx} (kx + 2) dx$$

but  $\neq f'(x)$

$$\therefore A = \frac{1}{k} \left[ x^2 e^{kx} \right]_0^2$$

$f(x)$

$$\therefore A = \frac{16}{k}$$

2-correct solns.

1+ simplifies integrand

$$A = \frac{1}{k} \int_0^2 f'(x) dx$$

$$\therefore \frac{16}{k} = \frac{1}{k} \left[ x^2 e^{kx} \right]_0^2$$

$$16 = [2^2 e^{2k} - 0]$$

$$16 = 4e^{2k}$$

$$4 = e^{2k}$$

$$\ln 4 = 2k$$

End of Exam

$$k = \frac{1}{2} \ln 4 \quad \text{or} \quad k = \ln \sqrt{4} = \ln 2$$